SECOND REPORT - Modelling frameworks (M1+M3)

Project Title - Efficiency and Bus Fleet Management: Strategic Communications and quantifying GHG and other co-benefits (#BBUS)

Submitted by - CEEW

June 2026

Contents

1. Context

CEEW analysis shows that more than 230 million users depend on bus systems for their daily mobility requirements. The majority of these users can be classified into a 'captive user group' having limited access to personal/private mobility modes. These populations comprise vulnerable road users such as women, the elderly, children and individuals from the economically weaker sections of society. Efficient public transport for these groups is not just about convenience, but a prerequisite for their socio-economic functioning. Indian cities are urbanising at a rapid pace, with around 600 million people expected to live in cities by 2036 (up from around 377 million in 2011). This urban expansion is generating enormous travel demand, but public bus systems are not keeping pace with it. Historically, Indian cities were dependent on public transport, but in recent decades, the trends have reversed in favour of private vehicles. Moreover, the supply of public transport is inequitable, as more than 60% of India's bus transport fleet is concentrated in just 9 metro cities. Overall, public transport, especially bus systems, are facing challenges such as limited supply, inconsistent service quality, poor investments, etc.

Thus, India's urban bus transport systems are overdue for an overhaul. A dual track strategy is needed: improve the narrative of bus systems beyond ridership and simplify investment frameworks to ensure capital flows where it's needed most. This can be done beyond anecdotal accounts to generate robust, data-driven evidence that captures the full spectrum of co-benefits offered by bus systems. A fresh, compelling narrative of quantified benefits of the bus deployment - climate, social, economic and health co-benefits - is urgently needed for India. Without such a comprehensive assessment, policy and funding decisions remain fragmented and reactive, leaving many regions dependent on private vehicles. India's transport sector is undergoing rapid change, with diverse regional trends. A cross-sectional evaluation across diverse geographies and bus systems is necessary. Interestingly, there is a growing portfolio of national and state-level bus funding schemes. A framework for quantifying the benefits of budgetary allocations for bus-based public transportation will help improve investments.

2. Objectives of the BBUS project

In the above context, this project will aim to generate evidence by developing a unique 4-1 co-benefits model covering socio-economic, health, GHG and resilience models to empower local/state/central governments to make data-driven decisions to increase the budgetary allocations and even attract climate funds and private sector investments for bus transport. The project scope encompasses the creation of a "Healthy Cities" Investment Case and a Social Return on Investment (SROI) framework, which will provide a multi-faceted assessment of returns on bus and e-bus investments. This will also help the decision-makers to improve their long-term strategies and demonstrate accountability. This framework is intended for use by the Ministry of Housing and Urban Affairs (MoHUA), NITI Aayog, and state-level transport departments/bus transport agencies to gauge the societal and environmental dividends of transit expenditure accurately.

The key objectives of the project remain:

- Develop an integrated '4-in-1 co-benefits model' to inform policy decisions and support the Monitoring, Reporting, and Verification (MRV) mechanisms of bus systems.
  - M1 - Identify socio-economic co-benefits, especially for women, low-income groups, and other vulnerable road users (VRUs).
  - M2 - Estimate the health and air quality benefits from increased bus usage and reduced reliance on private vehicles.
  - M3 - Quantify the greenhouse gas (GHG) mitigation potential of bus deployment.
  - M4 - Assess the resilience and adaptive benefits and impacts of robust bus-based public transport systems on the economy and physical infrastructure in cities.

- Easing access to climate, green funds and other potential avenues for bus and allied infrastructures
- Enhance understanding and belief in bus-based systems: Strategic communications of benefits amongst decision makers and government agencies

3. Socio-Economic Model (M1)

Indian cities are entering a decisive phase of public-transport investment. The PM e-Bus Sewa scheme, approved by the Union Cabinet in August 2023, commits approximately 10,000 electric buses across 169 eligible cities at a total estimated cost of ₹57,613 crore, with central support of ₹20,000 crore over ten years of operations. The scheme deliberately prioritises tier-2 and tier-3 cities that currently lack organised bus services, and runs on a Gross Cost Contract model. It is the most consequential central intervention in urban bus capacity since the Jawaharlal Nehru National Urban Renewal Mission, and the appraisal evidence it generates will shape the second-phase scheme already under planning.

Yet the analytical case for bus investment in India still rests almost entirely on fare-box revenue, operating-cost economics, and a narrow monetisation of in-vehicle travel-time savings. The wider returns — savings on household transport expenditure, expanded labour-market access (especially for women), improved educational and healthcare access for low-income households, road-network decongestion, and productivity gains from denser effective labour markets — are routinely acknowledged but rarely quantified. In their absence, conventional cost-benefit analysis systematically understates the social return to bus investment. The empirical gap is acute: a scoping review of twenty-seven studies identifies exactly one India-specific co-benefit study (Delhi non-motorised transport, 2022); the Indian evidence base is otherwise transferred from international settings, with transferability assumed rather than tested.

M1 model of the #BBUS study is designed to fill this gap. It quantifies the socio-economic co-benefits of bus-based public transport across two groups of Indian cities: Group A, where mature municipal bus systems already operate (Bengaluru, Jaipur, Bhubaneswar), and Group B, where the PM e-Bus Sewa rollout provides a phased deployment exploitable as a natural experiment (Haridwar, Gaya, Jodhpur, and others).

Three design choices structure the framework. First, the causal-identification strategy differs across the two groups: Group A admits only a rich-controls cross-sectional regression with sensitivity bounds for unobserved selection, while Group B supports a two-period difference-in-differences design that recovers the average treatment effect on the treated under parallel trends. Second, the equity dimension is built into the specification through a four-component Vulnerability Index (income, vehicle ownership, disability, elderly status) that enters every regression as both a main effect and an interaction with bus access, so the differential treatment effect along vulnerability is a coefficient of direct policy interest rather than a subgroup result. Third, decongestion (via a Marginal External Congestion Cost framework) and agglomeration (via a Graham/WebTAG effective-density framework) are estimated and reported separately from household-level co-benefits, following international convention, to avoid double-counting.

3.1 Literature Review

scope and approach

Twenty-seven studies were synthesised, prioritising literature from cities economically and demographically comparable to Indian urban contexts. The review covers the methodological evolution of transport appraisal, empirical estimates of socio-economic outcomes attributable to public-transport investments, and equity-oriented frameworks for capturing distributional co-benefits. A majority of the reviewed evidence (~59%) comes from Global South contexts; the balance (~41%) is from advanced economies where causal-identification methods and equity-focused appraisal are more developed.

Region	Representative cities	Share of reviewed studies
South and South-East Asia	Bogotá, Lagos, Dhaka, Lahore, Santiago	33%
East Asia (China, South Korea)	Shenyang, Hangzhou, Beijing	18%
India	Delhi(NMT Co-benefit study), Surat	8%
Australia / New Zealand	Melbourne(PT equity and health studies)	15%
North America / Europe	Toronto, Rome, Chicago, Miami, Austria, Atlanta	26%

Different Modelling Frameworks

Four candidate frameworks were considered. Each captures a distinct dimension of the bus-access–outcomes relationship and corresponds to a different analytical question that the M1 evidence base needs to answer. The four frameworks are not mutually exclusive; in principle, a co-benefit study can draw on all of them. The choice of which to centre is governed by the research design that the available data and the deployment context permit. This section discusses the conceptual and identification rationale for each.

1. Structural Equation Modelling(SEM)

SEM is a two layer statistical framework for systems in which several of the variables of interest cannot be measured directly. In Bus-based public transport, these typically include constructs such as perceived accessibility of the network, sense of personal safety on board, and overall satisfaction with the service– none of which can be read off of a single survey item, but each of which is reflected in the joint pattern of responses to a bundle of related questions. Perceived safety, for example, may be inferred from a rider's ratings of crowding, conductor and driver conduct, behaviour of co-passengers etc taken together.

SEM is not adopted for M1. The M1 module's objective is to quantify socio economic co-benefits – changes in transport expenditure, female employment, agglomeration gains etc. all of which are directly observable. The latent constructs that SEM is designed to recover such as perceived safety, service quality, driver attitudes –they address a different question.

2. Difference-in-Differences(DiD)

DiD estimates the effect of an intervention by comparing the before-and-after change in outcomes for a treated group with the before-and-after change for a comparable untreated group. The identifying assumption is parallel trends: absent the intervention, the two groups would have moved in step, so the control group's change pins down what the treated group would have done without it. The difference between the two changes recovers the Average Treatment Effect on the Treated (ATT).

The phased rollout of PM e-Bus Sewa maps cleanly onto this structure. Buses reach cities at different times, and within each rollout city only neighbourhoods inside the catchment of a new route are served. Comparing households in newly served catchments with households in similar unserved neighbourhoods, before and after the buses begin operating, isolates the effect of bus access on outcomes such as household transport expenditure, female labour-force participation, and access to jobs and services.

3. Multi-variate(cross sectional) Regression

Multivariate regression estimates how an outcome varies with several explanatory variables simultaneously, isolating the effect of the variable of interest — here, bus access — by holding other determinants of the outcome constant through a set of controls. The method delivers a credible estimate of the bus-access effect provided the controls are rich enough to account for the systematic differences between households that locate near bus corridors and those that do not.

In Group A cities (Bengaluru, Jaipur, Bhubaneswar), where mature bus networks have been in place for years, no pre-treatment baseline is observable and a before-after comparison is not feasible. The strategy is instead a cross-sectional regression of household outcomes — transport expenditure, female employment, time use, access to jobs and services — on a 500m walkshed indicator for bus access, conditional on household, neighbourhood, and locational controls.

4. Spatial/GIS Analysis

Spatial analysis is a class of methods for examining data that carry location as an attribute.In M1, spatial / GIS analysis serves three roles. It generates the bus-access treatment variable as a 500-metre network-based walkshed around operational or planned stops, using STU shapefiles, OpenStreetMap, and GTFS data. It supports the Group B matched control-zone design by identifying candidate control wards that match treatment wards on baseline density, road quality, and infrastructure. And it produces the descriptive outputs — coverage maps, gap analysis, hotspot maps, origin-destination flows — that inform the policy narrative without entering the estimating equations directly.

3.2 Research Gap

The M1 literature review identifies the following gaps, which this study seeks to bridge:

limited India-specific evidence : out of the 27 studies compiled for M1 only one is from India(Delhi NMT, 2022).The rest come from Latin America, East Asia, Europe, North America, Australia and other South Asian countries. Travel behaviour, income levels, vehicle mix, gender norms around mobility, the role of informal transport, and the institutional structure of urban bus services in Indian cities differ substantially.

Bus-specific evidence is thin even within the international literature. Most rigorous causal evaluations of public transport — including those in our reviewed set — are on metro, subway and rail systems

(Rome metro, Hangzhou subway, Chinese metro panel). The bus literature leans on descriptive accessibility indices, SEM and satisfaction surveys rather than causal designs.

Tier-2/tier-3 / smaller-city evidence is thin globally and domestically : Most rigorous evaluation work comes from megacities (Bogotá, Rome, Hangzhou, Melbourne, Chicago). Smaller-city deployments — which are exactly where most new bus investment in developing economies will land — are under-represented.

Socio-economic co-benefits beyond travel time savings are not monetised in Indian appraisal. Internationally, household co-benefits are monetised in some appraisal frameworks (UK WebTAG, parts of EU practice); in India, the case for bus investment rests on fare-box, operating economics and in-vehicle travel-time savings. Household transport savings, female labour-market access, and educational and healthcare access are routinely acknowledged but almost never quantified

Equity treated as a subgroup result rather than a structural parameter. Across both global and Indian bus-based PT studies, vulnerability is addressed through separate subgroup tabulations or layered equity indices. The differential treatment effect across vulnerability strata — whether buses benefit low-income, women, elderly and disabled riders more or less than the average rider — is rarely estimated as a coefficient of direct policy interest within the main estimating equation.

3.3 Selected instruments: household/individual level socio economic co-benefits

The selected design combines (i) **cross-sectional multivariate regression** for Group A (ii) two-period **2×2 difference-in-differences** for Group B, with treatment-control assignment by 500-metre walkshed and matched controls; and (iii) supporting spatial and descriptive analysis for both groups, where regression is infeasible. The bus-access measure — a binary indicator for residence within a 500-metre GIS walkshed of an operational or planned bus stop — is the primary independent variable. It is constructed from spatial data and is exogenous to individual-level behaviour.

All analyses — both Group A cross-sectional regressions and Group B difference-in-differences — are estimated city by city as the primary specification. The bus systems, labour markets, road networks, and household structures of the cities under study differ in ways that are likely to generate heterogeneous treatment effects: a 500-metre walkshed in Bengaluru plugs into a denser, IT-led labour market with a partial metro overlap, whereas the same nominal walkshed in Bhubaneswar or in a tier-2 PM eBus Sewa deployment city plugs into a smaller services-and-administration labour market with no rail competition and different fleet characteristics. Pooling under city fixed effects would absorb city-level mean unobservables but would impose a single average treatment effect across these heterogeneous deployments, masking variation that is itself policy-relevant — particularly so for Group B, where the PM eBus Sewa scheme is administered and evaluated city by city. Ultimately, the feasibility of city-wise estimation depends on the per-city sample realised in the field; where the achievable sample is too thin to support credible per-city inference, the pooled specification with city fixed effects becomes the primary report rather than the secondary.

City Group	Cities	Selected model	Identification rationale

Group A – Cities with existing buses	Bengaluru, Jaipur, Bhubaneshwar	Multivariate regression (cross-sectional)	Mature bus systems; no pre-bus baseline. Estimate the partial association between bus-access and outcomes under a selection-on-observables assumption, with rich controls and sensitivity bounds.
Group B– PM e-Bus Sewa deployment cities	Haridwar, Gaya, Jodhpur, etc.	Two-period 2×2 difference-in-differences	Phased deployment provides a natural experiment. Households within the planned walkshed are treated; matched households beyond the walkshed serve as controls. Identifies the ATT under parallel trends.

3.4 Outcome Variables

Outcomes are split between those modelled in regression and those handled descriptively. Regression is reserved for outcomes observed across the full sample, where bus *access* — a geographically defined dummy — is the regressor and identification is uncontaminated by behavioural self-selection. Outcomes meaningfully defined only on the sub-sample of actual bus users (value of travel time saved, educational and healthcare access) are reported descriptively: a regression on the user sub-sample would condition on an endogenous behavioural choice and would not recover a causal effect. The current list is provisional; additional outcome variables may be added and existing ones modified as the framework develops.

Outcome Variable	Analysis type	Description
Individual transport expenditure	Regression	Self-reported monthly spending on all transport modes (₹/month, continuous). Observed for the full sample; identification rests on the geographically defined bus-access dummy.
Employment status	Regression	Binary indicator of current employment. Observed for the full sample; captures the labour-market participation response to expanded mobility.
Female employment status	Regression	Binary employment indicator estimated on the women-only sub-sample. Isolates the gendered labour-supply response while retaining bus access as the (exogenous) regressor.
Women's trip frequency	Regression	Count of daily out-of-home trips by women in households without private vehicle access (captive users). The

		captive-user restriction is on a *household asset* variable, not on bus usage, so identification is preserved.
Value of travel time saved	Descriptive	Monetised travel-time savings constructed post-survey for bus users, using reported commute times and a state counterfactual mode. Because the quantity is defined only on bus users, any regression would condition on an endogenous behavioural choice (bus usage) and would not recover a causal effect.
Educational access for children/youth	Descriptive	School/college days missed due to transport, and stated counterfactuals on institution choice. The informative variation lies among households whose children use the bus, so regression would condition on self-selected bus use.
Healthcare access	Descriptive	Healthcare visits delayed or skipped due to transport, and mode used for healthcare visits. As with the educational outcome, the meaningful sub-sample is bus-using households, making regression vulnerable to self-selection bias.

3.5 Explanatory variable: Bus Access

Bus access enters the specification as a binary indicator equal to 1 if the household lies within a 500-metre walkshed of an operational (Group A) or planned (Group B) bus stop, and 0 otherwise. The walkshed is constructed from spatial data — STU route shapefiles, OpenStreetMap, and GTFS feeds where available — using a network-based rather than Euclidean buffer, so that the measure reflects the road and footpath geometry pedestrians actually traverse. Walking distance to the nearest stop is also collected directly in the survey (in minutes and metres) and cross-validated against the GIS measure.

3.6 Vulnerability Index(VI)

The Vulnerability Index (VI) is a normalised composite intended to capture multiple dimensions of socio-economic vulnerability that condition the marginal benefit a household derives from bus access. It enters the regressions as a standalone regressor — capturing baseline disadvantage of vulnerable road users (VRUs) — and as an interaction term with bus-access — capturing the differential treatment effect across vulnerability strata.

Component	Description
*Income	Household monthly income min-max scaled and reversed. closer to 1 means most vulnerable
vehicle ownership	binary variable, = 0 if household own vehicle ,=1 if owns no vehicle
**disability status	binary variable, =0 if no disability , =1 if disabled

Elderly	binary variable, =0 if age < 60 , =1 if age >=60

* income thresholds should be decided.
**can use some benchmark or statutory basis to attribute disability.

We construct the index as follows. 1 is the most vulnerable and 0 is the least vulnerable.
VI = (income + vehicle ownership + disability status + elderly)/4

An equal-weighting scheme is the default. The choice is normative — it assigns equal importance to each dimension of vulnerability — and follows the convention used in widely-cited composite indices in development policy, most prominently UNDP's Human Development Index.

Gender is not included in the VI. Classifying all women as vulnerable regardless of their income or vehicle access would obscure the variation among women, which is the group we most want to study. Gender is instead used as a control variable in the regressions.

3.7 Group A cities: Cross sectional specification

The estimating equation is:

$$Y\_ij = \beta_0 + \beta_1 \cdot bus\_access\_j + \beta_2 \cdot VI\_ij + \beta_3 \cdot (bus\_access\_j \times VI\_ij) + \beta_4 \cdot X\_ij + \beta_5 \cdot \Gamma\_j + \varepsilon\_ij$$

where i indexes individuals, j indexes wards/neighbourhoods, X_ij is a vector of individual and household controls, Γ_j is a vector of neighbourhood/ward controls, and ε_ij is an idiosyncratic error term. Standard errors are clustered at the ward level.

interpretation of coefficients is as follows:

Coefficient	Interpretation	Policy role
β_1	Conditional difference in outcome between individuals with bus access and without, at VI = 0 (the reference, least-vulnerable group).	Headline policy number — the unconditional treatment effect for the reference group.
β_2	Conditional effect of vulnerability on the outcome, independent of bus access (evaluated at bus_access = 0).	Captures the baseline disadvantage of VRUs.
β_3	Differential treatment effect of bus access for a one-unit increase in VI. Statistically, this is the heterogeneity-in-treatment-effect parameter	Equity coefficient. A significant positive β_3 is the core equity argument for bus investment

3.8 Group B cities: 2x2 DiD specification

The estimating equation is:

$$Y\_ijt = \beta_0 + \beta_1 \cdot bus\_access\_j + \beta_2 \cdot (bus\_access\_j \times post\_t) + \beta_3 \cdot (bus\_access\_j \times VI\_ij \times post\_t) + \beta_4 \cdot VI\_ij + \beta_5 \cdot X\_ijt + \beta_6 \cdot \Gamma\_j + \lambda\_t + \varepsilon\_ijt$$

where t indexes survey round (pre/post), post_t is an indicator for the post-deployment wave, and λ_t is a time fixed effect. The coefficient of policy interest is β_2, which identifies the ATT(average treatment effect on the treated) under parallel trends and SUTVA.

the main coefficients are interpreted as follows:

Coefficient	Interpretation	Policy role
β_2	Average treatment effect on the treated (ATT) under parallel trends: the mean change in YY for individuals inside the 500m walkshed that is attributable to the bus deployment, net of the common time trend. Identified as the difference-in-differences between treated and control units across the two waves.	The headline causal estimate — the answer to "did the deployment work, on average?" Reported as the primary impact figure
β_3	Differential ATT by vulnerability: the extent to which the treatment effect varies with the vulnerability index(VI).A positive β_3 means more vulnerable individuals gained more from the deployment than less vulnerable individuals inside the same walkshed.	The equity coefficient — the answer to "did the deployment work *for the people it was meant to serve?*" Distinguishes broadly progressive deployments from those that merely raised the average

Controls: These are common to both Group A and group B specifications.

individual controls: household size,gender, marital status,household income etc..

trip/service related controls: commute distance, trip frequency etc..

ward/neighbourhood controls: population density, road density etc..

*we can add more controls. The main purpose of adding controls is to remove the risk of endogeneity. which means there is an unobserved variable which covaries with both bus access and our outcome variable. technically endogeneity is represented as $COV(X,\varepsilon) \neq 0$ where X is a regressor and ε is the error term.

** We will drop the gender specific controls when the dependent variable is gender oriented.otherwise we will break some assumptions(Rank(X) = no of columns) needed for ols estimation.

3.9 Wider economic benefits (system vs city)

Above sections estimate household/Individual-level co-benefits — savings on transport expenditure, employment uptake, women's mobility, and so on — that accrue to identified individuals on identified trips. These are the marginal-utility benefits of expanded mobility, captured in surveys and disciplined by causal-identification arguments. A second class of benefits accrues at the level of the urban economy as a whole, beyond the bus rider: lower congestion costs on shared road infrastructure and higher productivity from denser, better-connected labour markets. These benefits are conventionally bundled together as “Wider Economic Benefits”(WEB) and are reported separately from individual co-benefits to avoid double counting and to follow the conventions established.

Two channels are modelled in M1: decongestion (a road-network externality) and agglomeration(a labour market productivity externality). Both are sourced from frameworks developed in the academic literature.

Decongestion Benefits—MECC × ΔVKT

The decongestion benefit of bus deployment is monetised using the Marginal External Congestion Cost framework. The benefit accounting combines the unit-MECC monetisation framework of Aftabuzzaman, Currie & Sarvi (2010) with the PCE-km convention of IRC:106 (Indian Roads Congress, 1990), following standard Indian transport-appraisal practice. The conceptual point is straightforward: a private vehicle journey imposes a delay externality on every other vehicle on the road network, and a public-transport service that diverts that journey removes the externality. The monetised value of the removed externality is the social benefit of decongestion. The formal derivation from a BPR(Bureau of public roads)-style congestion function is given in Appendix B.

Inputs:

1. Counterfactual mode (the mode the bus rider would otherwise have used) — from stated-preference questions in the survey.
2. Trip length by counterfactual vehicle type.
3. Vehicle occupancy by counterfactual vehicle type.
4. Total daily ridership — from Automatic Fare Collection Systems and Gross Cost Contract logs.
5. Route and network length.
6. Passenger Car Equivalent (PCE) values — from IRC:106 (Indian Roads Congress, 1990).
7. MECC unit values per PCE-kilometre, sourced from Sen, Tiwari, and Upadhyay (2010) and updated where local administrative data permit.

measurement:

Stage	Calculation
1. Diverted trips	Let “d” be the share of riders who would otherwise have used a car, two-wheeler, or other private mode. Diverted trips = total ridership × d. Computed separately by counterfactual mode.
2. Gross VKT	(Diverted trips ÷ average vehicle occupancy) × average trip length. Computed separately for each counterfactual vehicle type.
3.Net PCE-Km	Σ (VKT × IRC:106 PCE factor) − total bus PCE-km. Subtracting the PCE footprint of the bus service itself yields the net change in PCE-km on the network.
4. Monetisation	Annual decongestion benefit = Net PCE-km × 365 × MECC unit value.

Interpretation of results by city group. Group A and Group B operate under different identification regimes, and the decongestion estimates carry different epistemic weight.

For Group A, no clean before-and-after comparison is feasible: the bus systems have been in place for years and no pre-bus baseline exists. The counterfactual mode share “d” is drawn from stated-preference questions, and the resulting estimate is interpreted as the congestion cost currently averted by the existing network — a descriptive quantity rather than a causal one.

For Group B, the two-wave survey supplies a clean counterfactual. The estimate is reported as the *causal reduction* in congestion cost attributable to deployment.

Agglomeration economies–Graham / WebTAG effective-density framework

Beyond the direct, individual travel-time savings already captured in standard CBA, transport investments raise aggregate productivity by increasing the effective economic size of the city — the labour market accessible from each location, weighted by travel cost. The empirical translation into transport appraisal — the effective-density framework adopted in this study — follows Graham (2007) and Venables (2007), and forms the basis of the UK Department for Transport's WebTAG guidance on wider economic impacts. The formal production-function micro-foundation and the identification problem for the agglomeration elasticity ρ are developed in Appendix C.

The primary channels through which transport investment increases productivity are:

1. **Matching.** A thicker labour market improves the quality of worker–firm pairings: with more workers and firms within reachable commute distance, each side searches across a wider pool and assignment moves closer to its productive optimum (Diamond 1982; Helsley and Strange 1990). Better bus access enlarges this reachable pool directly.
2. **Knowledge spillovers.** Tacit knowledge — skills, techniques, and ideas not codified in manuals — diffuses through face-to-face contact and worker movement between firms (Jacobs 1969; Glaeser et al. 1992). The spillover decays sharply with distance, within one to five kilometres (Rosenthal and Strange 2008). Bus access enlarges the population that can reach knowledge-dense employment clusters.
3. **Input sharing.** Larger markets support finer division of labour (Smith 1776; Stigler 1951). Intermediate inputs and specialised producer services — accountants, engineers, logistics providers — are cheaper and more varied in dense areas, because suppliers spread fixed costs over a larger client base (Holmes 1999). Transport supports the worker movement that knits the local supplier–firm network together.

Measurement

Stage	Step	Specification
1.	Define Zones	Divide each city into ward level zones. Build the employment and Gross Value Added(GVA) panel from the economic census and PLFS.
2.	Effective Density	$U_i = \Sigma_j E_j \cdot f(d_{ij})$, where E_j is jobs in ward j, d_{ij} is generalised travel cost from j to i, and $f(\cdot)$ is a distance-decay function (negative-exponential or power form; sensitivity reported).

3.	Change from investment	$\Delta U_i / U_i$ — bus deployment lowers d_{ij} on served corridors, raising U_i. On unserved corridors, U_i is unchanged.
4.	monetise	Welfare uplift in ward i: $\rho \cdot (\Delta U_i / U_i) \cdot GVA_i$, where ρ is the agglomeration elasticity — the percentage change in productivity associated with a one-percent change in effective density.

Aggregation across wards: **$\Delta WEB = \Sigma_i \rho \cdot (\Delta U_i / U_i) \cdot GVA_i$.** This metric gives the agglomeration benefits out of a transport investment.

4. Health Co-benefits Model (M2)

Bus-based public transport is not only a mobility intervention; it is also a public health intervention. By improving bus access and service quality, cities can reduce dependence on private and intermediate motorised modes, improve daily walking through access and egress trips, reduce exposure to transport-related air pollution, lower road injury exposure, and reduce commute-related stress. Model 2 of the BBUS framework is designed to estimate these health and air-quality co-benefits and make them visible for public investment decisions.

The M2 model will quantify the health value of increased bus usage and reduced reliance on private vehicles across Indian cities. It will generate evidence for decision-makers by linking changes in travel behaviour with changes in active travel, air pollution exposure, road safety exposure, commute stress, and health-burden outcomes. The model will also contribute the health component to the wider 4-in-1 BBUS co-benefits framework and Social Return on Investment assessment.

4.1 Literature Review

The literature review for M2 synthesises evidence from transport-health impact assessment studies, public transport and physical activity studies, air pollution exposure studies, road safety studies, commute stress studies, and public transport evaluation studies. The reviewed evidence shows that transport interventions influence health through multiple pathways rather than a single outcome.

Studies using transport health-impact models show that changes in travel behaviour can affect physical activity, air pollution exposure, road traffic injuries, and disease burden. Public transport and physical activity studies show that transit users often gain daily walking through access and egress trips. Commute stress studies show that travel mode, waiting time, crowding, reliability, comfort, perceived safety, and travel-time uncertainty influence stress and well-being. Public transport evaluation studies also show that bus access can influence social inclusion, independent travel, road safety, active travel, and economic value.

The reviewed literature is geographically concentrated in North America and Europe/UK, with limited evidence from Asia and very limited direct evidence from India. This makes it necessary to adapt globally established transport-health methods to the Indian bus context rather than directly transferring results from international studies.

Paper / report name	**Inference for BBUS M2**
Relation Between Higher Physical Activity and Public Transit Use, USA	Public transit use is linked with higher physical activity, supporting the active travel pathway through walking to and from bus stops.
Walking to Public Transit: Steps to Help Meet Physical Activity Recommendations, USA	Transit users gain walking time through access and egress trips, which can be used to estimate physical activity benefits from bus use.
Health Cobenefits and Transportation-Related Reductions in Greenhouse Gas Emissions in the San Francisco Bay Area, USA	Provides the strongest model backbone for linking transport mode shift with physical activity, air pollution, road injury and disease-burden outcomes.
Air Pollution as a Risk Factor in Health Impact Assessments of a Travel Mode Shift Towards Cycling, Global review	Shows how air pollution exposure is estimated in transport health-impact assessments, supporting the pollution exposure pathway in M2.
On the Buses: A Mixed-method Evaluation of the Impact of Free Bus Travel for Young People, UK	Shows that bus-related interventions can affect travel behaviour, active travel, safety, social inclusion, well-being and economic value.

Health Impact Assessment of Transport Initiatives: A Guide, Scotland / UK	Supports Health Impact Assessment as the umbrella framework for linking transport proposals with physical activity, pollution, injuries, mental health and equity.
Navigating Modal Shift: Exploring Public Transport Preferences in Coimbra, Portugal	Highlights the role of service quality, reliability, accessibility and user perception in encouraging shift towards public transport.
Towards a Low Emission Transport System: Evaluating the Public Health and Environmental Benefits, Iran	Shows how transport strategies can generate public health and environmental co-benefits through mode shift and emission reduction.
Public Transport Users versus Private Vehicle Users: Differences about Quality of Service, Satisfaction and Attitudes Toward Public Transport in Madrid, Spain	Useful for comparing public transport users and private vehicle users, and for analysing satisfaction, service quality and willingness to shift.
Am Stressed, Must Travel: The Relationship Between Mode Choice and Commuting Stress, Canada	Supports the commute stress pathway by showing how stress varies by travel mode, comfort, safety, commute time and service conditions.
Relation Between Higher Physical Activity and Public Transit Use, USA	Duplicate record in the literature sheet; should be counted as a reviewed record but not as a separate unique study.

4.2 Research Gap

The M2 literature review identifies three key research gaps.

Gap 1: Limited India-specific evidence. Most existing studies are based on cities outside India, where travel behaviour, public transport quality, pedestrian infrastructure, vehicle mix, pollution exposure, road safety risks and service reliability differ from Indian urban conditions. This limits direct transferability to Indian bus systems.

Gap 2: Limited bus-specific evidence for the BBUS policy question. Existing multi-pathway studies often assess broad transport scenarios, active travel, compact-city planning or fare policies. They do not directly examine how improved bus access and service quality can shift users from two-wheelers, cars, autos and informal modes to buses, and how this shift affects health outcomes in the Indian context.

Gap 3: Limited conversion of health benefits into investment indicators. Health benefits such as avoided disease burden, reduced injuries, lower commute stress and potential health-care cost savings are often discussed, but they are rarely converted into standardised indicators that can support bus planning, public investment appraisal or the wider Social Return on Investment framework.

4.3 M2 Framework

The M2 framework will assess four health pathways.

The first pathway is active travel. Bus users generally walk to the boarding stop and from the alighting stop to their final destination. This access and egress walking can increase daily physical activity.

The second pathway is air pollution exposure. If users shift from two-wheelers, cars, autos or other motorised modes to buses, private and intermediate vehicle travel can reduce. This may reduce exposure to pollutants such as PM2.5 and NO2.

The third pathway is road injury exposure. Shifting from higher-risk private or intermediate modes to buses may reduce road injury exposure. However, the model will also examine whether walking to and from bus stops increases pedestrian exposure in unsafe road environments.

The fourth pathway is commute stress and well-being. Reliable, affordable, safe and comfortable bus services can reduce waiting stress, crowding stress, safety-related stress, travel-time uncertainty and affordability pressure.

Together, these four pathways will allow M2 to estimate the health value of bus systems beyond ridership, fare revenue, or travel-time savings.

4.4 Methodological Approach

M2 will use an adapted transport Health Impact Assessment framework supported by primary surveys, GIS exposure mapping, health-burden data, air pollution data, road safety data and bus system data.

For active travel, the model will estimate walking generated by bus access and egress. For air pollution, it will estimate reduced exposure and emissions from mode shift away from private and intermediate motorised modes. For road safety, it will compare reduced private-mode exposure with added pedestrian access exposure near bus stops. For commute stress, it will use survey-based indicators covering waiting time, reliability, crowding, safety, comfort, fare burden, heat exposure and travel-time uncertainty.

The framework will not directly transfer international results. Instead, it will adapt globally established methods and calibrate them using Indian city-level data collected under BBUS.

4.5 Candidate Modelling Frameworks Considered

The M2 literature review shows that no single model can capture all health co-benefits of bus-based public transport. Different methods are useful for different parts of the health question. Therefore, M2 will use an adapted Transport Health Impact Assessment framework as the core approach, supported by statistical, behavioural and spatial methods.

Candidate modelling framework	**What it can measure**	**Limitation**
Transport Health Impact Assessment	Links bus interventions with health outcomes such as physical activity, air pollution exposure, road safety and well-being	Requires reliable local exposure and health data
ITHIM-style comparative risk assessment	Estimates health impacts from changes in active travel, air pollution and road injury exposure	Needs local calibration; does not fully capture commute stress or service perception
Public transport walking / active travel model	Estimates walking generated by access and egress trips to bus stops	Covers only one pathway
Ordered Logit / Ordered Probit	Analyses ordered survey responses such as stress level, safety perception, comfort, satisfaction and willingness to use bus	Does not estimate full health burden by itself
Structural Equation Modelling / Generalised SEM	Tests latent constructs such as service quality, perceived safety, affordability pressure, commute stress and bus adoption intention	Not suitable as the main model for DALYs, pollution or injury burden
Cross-sectional regression with rich controls	Compares bus users, non-users and users of private/intermediate modes while controlling for socio-economic and trip characteristics	Cannot fully prove causality because it uses one-time data
Difference-in-Differenc es / longitudinal panel model	Compares before-after changes between people or areas exposed to bus deployment and comparable control groups	Requires comparable treatment and control groups and follow-up data

GIS-based accessibility and exposure analysis	Maps bus stop access, walking catchments, route coverage, pollution exposure and crash hotspots	Needs accurate GIS and exposure datasets
Cost-benefit / SROI health valuation	Converts selected health outputs into economic or investment-relevant indicators	Depends on valuation assumptions and should not be used alone

Based on this comparison, M2 will use an adapted Transport Health Impact Assessment framework as the main model. The ITHIM-style approach will support quantification of physical activity, air pollution and road injury pathways. SEM and ordered response models will support commute stress, perceived safety, service quality and bus adoption analysis. GIS will provide the spatial exposure layer, while regression and longitudinal methods will support Group A and Group B city designs respectively.

4.6 Available Tools for M2 Assessment

Several directly available tools can support the M2 assessment. ITHIM can be used as the main health-impact tool for estimating physical activity, air pollution and road injury pathways. WHO HEAT can support the estimation of health and economic benefits from walking to and from bus stops. WHO AirQ+ and BenMAP-CE can support air-pollution health-impact estimation and valuation. iSThAT can be used as a reference tool for sustainable transport health and economic benefits. QGIS can support bus-stop catchment, walking-access, pollution-exposure and crash-hotspot mapping. iRAP/ViDA can support road-safety assessment around bus stops and access routes. R or Stata can be used for ordered response models, regression and SEM/GSEM to analyse commute stress, perceived safety, service quality and willingness to shift to buses.

4.7 Group-wise Application

For Group A cities, where bus systems already operate, M2 will use a cross-sectional design. The model will compare regular bus users, occasional bus users, non-users, and users of two-wheelers, cars, autos and informal modes. The purpose is to estimate the current health value of existing bus systems and identify which service-quality, access, safety and affordability factors are linked with health and well-being outcomes.

For Group B cities, where bus deployment is expected, M2 will use an ex-ante and longitudinal design. Before bus operations begin, the model will estimate expected health benefits using planned routes, planned bus stop access, current mode, trip distance, stated willingness to shift, expected waiting time, expected fare, expected safety and expected reliability. After operations stabilise, the same respondents can be followed up to validate actual changes in walking, mode shift, exposure, safety perception and commute stress.

This group-wise design allows M2 to support both current assessment of existing bus systems and pre-investment appraisal of future bus deployment.

4.8 Data Requirements

M2 will require six data blocks.

The first is primary commuter survey data, including current mode, alternative mode if bus is unavailable, trip purpose, trip distance, travel time, travel cost, access walking, egress walking, waiting time, commute stress, perceived safety, self-reported health, crash or near-miss experience, and socio-economic characteristics.

The second is GIS data, including home and destination locations, bus stop locations, route alignment, walking catchments, access distance, pollution exposure layers and crash hotspot locations.

The third is bus system data, including routes, stops, frequency, fare, ridership, service reliability, crowding, fleet type and operating schedule.

The fourth is air pollution data, including PM2.5, PM10, NO2, route-level exposure, city-level exposure and emission factors.

The fifth is road safety data, including crashes, injuries, fatalities, victim mode, striking vehicle mode, road type and blackspot locations.

The sixth is health-burden data, including DALYs, deaths, Years of Life Lost and Years Lived with Disability by cause, age, sex and location.

4.9 Expected Outputs

M2 will produce pathway-wise and city-wise health co-benefit indicators.

The expected outputs include additional walking generated by bus use, reduction in air pollution exposure, reduction in road injury exposure, reduction in high-stress commuters, DALYs averted, injuries avoided, potential hospitalisation or treatment cost savings, and health co-benefits for vulnerable groups.

The outputs will be disaggregated by gender, income group, age group, disability status, vehicle ownership, ward and city. This will help identify who benefits most from bus systems and where additional investment is required.

4.10 Policy and Investment Relevance

M2 will help decision-makers understand the public health value of bus investment. It will show how buses contribute to cleaner air, safer mobility, active travel, lower commute stress and healthier cities.

For local governments, M2 can identify bus stops, corridors and wards where better access, footpaths, crossings, lighting, service reliability or frequency improvements are needed. For state transport departments, it can compare health benefits across cities and user groups. For central agencies and funders, it can provide standardised health indicators that feed into the wider BBUS 4-in-1 co-benefits model.

M2 will not independently create a finance proposal. Instead, it will provide the health component of the wider Social Return on Investment framework. Its outputs, such as DALYs averted, injuries avoided, high-stress commuters reduced and potential health-cost savings, will be combined with socio-economic, GHG and resilience benefits from the other BBUS models to strengthen the overall investment case for bus-based public transport.

4. GHG benefit model (M3)-

The transport sector is a major and rapidly growing source of greenhouse gas (GHG) emissions, making accurate measurement and monitoring essential for achieving climate and net-zero goals. At the city level, effective climate action requires robust tools to quantify emissions, identify mitigation opportunities, and track progress over time. However, urban transport emissions accounting in India is often constrained by the lack of standardised methodologies and reliable activity data, compounded by the country's diverse vehicle mix and complex travel patterns.

To address these challenges, the M3 Model has been developed as a bottom-up emissions assessment framework tailored to Indian cities to understand the GHG mitigation from Public transport (buses). Aligned with internationally recognised methodologies such as ITF-OECD and UNFCCC, the model integrates local travel behaviour, transport activity, and operational characteristics to estimate emissions and mitigation impacts more accurately. Using datasets such as travel diary surveys, Origin-Destination surveys, traffic counts, ridership records, and public transport operational data, the model quantifies avoided Vehicle Kilometres Travelled (VKT) and emissions reductions resulting from public transport interventions and modal shifts.

The framework incorporates city-specific emission factors based on fuel consumption patterns, vehicle technologies, fuel characteristics, electricity grid emission intensities, and electric bus charging profiles, ensuring greater accuracy than conventional top-down approaches. It also enables future scenario analysis by evaluating the impacts of strategies such as public transport expansion, fleet electrification, renewable energy integration, and operational efficiency improvements. By generating transparent and policy-relevant emissions inventories, the M3 model supports evidence-based decision-making, helps cities monitor climate targets, and strengthens the assessment of urban transport contributions toward India's NDCs and long-term decarbonisation goals.

4.1 Literature Review -

This literature review synthesises evidence from twelve (n=12) peer-reviewed studies, international frameworks, and applied modelling exercises to inform the design and calibration of Module 3, the GHG Emission and Mitigation Co-Benefit (BBUS) model.

a. Geographic coverage and Source distribution -

The review draws on twelve studies selected through a systematic mapping exercise, with geographic coverage deliberately weighted toward India, the primary deployment context of the project, while retaining sufficient international evidence to ensure methodological rigour and NDC alignment.

Table 1 - 58% of the India-focused literature helps to understand the context-specific GHG mitigation Model

Geographic Region	Number of studies	Share	Analytical Focus
India (National level study, City level study, Multi-city study)	7	58%	Bottom-up GHG, modal shift, BRT co-benefits
Global/ OECD framework	4	34%	WTW/TTW boundaries, NDC reporting standards, Cross-country model Validation and comparison
Global South (Case Country-Africa)	1	8%	Developing country analogy (similar economic Region analysis)

Source- CEEW analysis

b. Different Modelling Frameworks and Analytical parameters -

The literature converges on five modelling frameworks that together constitute the methodological architecture for a credible, policy-relevant BBUS project M3 Model. The table below synthesises each framework's core components and its specific application within this project.

Table 2 -42% of the reviewed literature suggested the ASIF framework for the GHG benefit model

Framework / Model	Core Components	Project Applications / Remarks
ASIF framework (Kumar et al. 2022)	Activity × Structure × Intensity × Fuel — foundational decomposition approach used by TERI, IEA, ITF, ICCT	Core architecture for the BBUS GHG model; links passenger-km, modal share, energy intensity and emission factors in a single accounting identity.
ITF WTW / TTW (LCA)	Well-to-Wheel and Tank-to-Wheel lifecycle boundaries; Scope 1/2/3; load-factor adjusted emission factors; grid carbon intensity; vehicle lifetime GHG	Both TTW and WTW metrics should be reported for UNFCCC/NDC compliance; it aligns BBUS with international accounting norms
Modal Shift / Substitution (Wani and Mohan 2025)	India EFs: 2-Wheeler = 0.051 $kgCO_2$/km; Bus = 0.015 $kgCO_2$/pax-km; net saving = 0.036 $kgCO_2$/pax-km (2W→Bus); validated via travel diary surveys	Core BBUS impact calculation — requires a before-and-after survey of new bus users to establish actual modal displacement
ASI Framework (Avoid–Shift–Improve) (Manan et al. 2022)	Three-pillar mitigation taxonomy: AVOID (trip reduction via TOD), SHIFT (modal switch to bus/PT), IMPROVE (fleet electrification); all three reported separately for NDCs	BBUS is primarily a SHIFT intervention with secondary IMPROVE benefits from fleet renewal/electrification; AVOID lens is optional for TOD-linked projects

ExSS / Scenario Analysis (Pathak and Shukla 2016)	Extended Snapshot model; compares BAU, Moderate and Ambitious scenarios; VKT × EF by mode per year; incorporates grid decarbonisation trajectory (2025–2035)	Long-term projections for BBUS should model three scenario pathways against baseline; the Surat 2025 study provides direct precedent with 11% city-level reduction under the PT-focused scenario
IPCC Tier 2 (Mobile Combustion) (Gajjar, Sheikh, and India GHG program 2015)	Diesel bus emission factor: 74.1 $kgCO_2/GJ$; CH_4 & N_2O by technology and fuel; GWP: CH_4=28, N_2O=265 CO_2e; India-specific EFs preferred over Tier 1 defaults	Required for UNFCCC-compliant GHG inventory; India-specific emission factors from MoRTH / national inventory should be used wherever available

Source- CEEW analysis

c. Different Tools and analytical framework -

Five different tools both national and international have been studied to understand how existing models work and what tools will be the best fit in BBUS project scope.

Table 3- Different tools available in secondary literature

Tool	**Main component of the tools**	**Remarks**
IFI TWG Harmonised tool (UNFCC)	Project-level GHG accounting: with-project vs. BAU baseline using ASIF-aligned emission factors	Project-level corridor or fleet; **less suited to city-wide multi-route network.**
ITF-OECD India LCA tool (ITF/NDCTIA , India specific)	Full life-cycle assessment (LCA) of passenger transport modes across vehicle manufacturing, fuel production, operation, and infrastructure construction	City-scale urban passenger transport — designed for Indian cities. Good for inter-modal comparison within one city context. **But best for - Comparinging ICE vs EV buses, Battery technology assessment , Fleet procurement decisions and fleet decarbonisation analysis.**
Green House Gas protocol- GHG calculation Tool (WRI/WBCSD corporate & City tools)	Corporate/city inventory across Scope 1, 2 & 3 using fuel-based, distance-based, or spend-based methods. Measures absolute organisational emissions.	Cannot model counterfactual — measures what IS, not what WOULD BE. **Not designed for additionality.**
UNFCCC GHG CRT tool (UNFCC)	National inventory for governments under Paris Agreement ETF. Reports annual GHG by IPCC sector (energy, transport, LULUCF, waste) using Tier 1–3 methods.	Requires analyst to construct greenfield counterfactual; guidance thin for zero-transit contexts.**Baseline assumption drives entire result;no methodology for 2W / Informal transit substitution.**
India GHG program Tool (WRI India/ TERI	GHG Protocol-aligned corporate inventory tool adapted for Indian	Only supports current inventory calculation for existing transport activity. **Cannot model**

/ CII)	businesses. Develops India-specific emission factors for road, rail, and air transport (Scope 3).	**bus absent scenario.**

Source- CEEW analysis

d. Research Gap in existing methods -

1. Most Indian studies use **Static average load factors. Dynamic load factor modelling, reflecting the peak and off-peak variation,** corridor-specific and seasonal demand fluctuations.
2. No reviewed study adequately **quantifies the emission impact of shifts away from informal para-transit (shared autos, tempos) toward formal bus services**. This gap is significant in the Indian context, where para-transit accounts for a substantial share of urban motorised trips.
3. Existing GHG tools are not designed for **city-level transport emissions assessment under Indian mixed-traffic conditions**. Among these tools ITF -OECD India LCA tool will be the most suitable one for the city specific analysis but the scope of the tool slightly differs from the scope of the study.
4. Most frameworks **cannot quantify emissions reductions from modal shifts** resulting from public transport interventions.
5. Current approaches rely heavily on **top-down fuel consumption data rather than actual travel behaviour and VKT changes**.
6. Existing models largely focus on CO_2 emissions and provide **limited assessment of air pollutant co-benefits ($PM_{2.5}$, PM_{10}, NO_X)**.
7. There is no widely adopted framework that combines **modal shift, avoided VKT, electrification, life-cycle emissions, and scenario analysis** within a single city-level assessment tool.

e. Rationale for the Selected Approach

To achieve the Project objectives, a bottom-up accounting approach based on the ASIF (Activity–Structure–Intensity–Fuel) framework has been adopted and integrated with a Well-to-Wheel (WTW) emission assessment methodology derived from the GREET framework.

This combined approach was selected because it provides a robust and transparent mechanism for estimating emissions from urban transport systems while capturing the specific pathways through which public transport interventions influence GHG emissions. Unlike top-down approaches that rely on aggregate fuel consumption data, the selected methodology enables the assessment of actual travel behaviour, modal shifts, vehicle performance, and fuel characteristics at the city level.

Framework -

Modified Formula for calculating existing GHG calculation (CO2 /Year)

GHG (t) =GHG(Tank to Wheel)+GHG (Well to Tank)+GHG (Grid Emission) +GHG (Battery Manufacturing)

i) Application of the ASIF Framework-The ASIF framework is widely recognised and used in transport emission studies globally due to its ability to link transport demand with energy use and emissions. The framework captures four key determinants of transport emissions:

- **Activity (A):** The volume of passenger travel undertaken within a city.

- **Structure (S):** The distribution of travel across different transport modes.
- **Intensity (I):** The amount of energy required to move passengers.
- **Fuel (F):** The carbon intensity of the energy source used.

The framework is particularly relevant for assessing bus-based public transport interventions because such interventions directly influence multiple components of the ASIF framework. The introduction or expansion of bus services can encourage a shift from private motorised modes to public transport, thereby altering the modal structure of travel. Improvements in vehicle occupancy and operational efficiency can reduce energy intensity per passenger-kilometre, while the adoption of cleaner fuels or electric buses can further reduce emissions associated with the fuel component.

By explicitly accounting for these relationships, the ASIF framework enables a comprehensive assessment of the mechanisms through which public transport contributes to emission reductions.

ii) Adoption of a Bottom-Up Emission Accounting Approach- The study adopts a bottom-up methodology using city-specific operational and travel data. Emissions are estimated based on vehicle kilometres travelled (VKT), vehicle fuel consumption, electricity consumption, fleet characteristics, passenger demand, and modal shift information collected through surveys and secondary data sources.

This approach offers several advantages for city-level analysis:

- It reflects local travel patterns and transport system characteristics.
- It allows estimation of emissions at the route, fleet, and city levels.
- It captures the impact of specific policy interventions and service improvements.
- It supports comparison between existing and future transport scenarios.
- It provides a transparent basis for estimating emissions reductions attributable to public transport investments.

Given the diversity of urban transport conditions across Indian cities, a bottom-up approach is considered more appropriate than applying generic national-level emission estimates.

iii) Integration of Well-to-Wheel Emissions-To ensure a comprehensive accounting of transport-related emissions, the methodology incorporates a Well-to-Wheel (WTW) perspective. This includes both direct emissions from vehicle operation and indirect emissions associated with fuel production and energy generation.

For internal combustion engine (ICE) vehicles, emissions are calculated using two components:

- **Tank-to-Wheel (TTW):** Emissions generated during vehicle operation through fuel combustion.
- **Well-to-Tank (WTT):** Emissions associated with fuel extraction, refining, processing, and distribution.

Tailpipe (ICE)

$GHG\ (TTW) = i\sum\{VKTi(Km/Year) \times FCi(l/Km) x\ (TTW)EFi(Kg/l)\}$

$GHG\ (TTW) = Kg/\ Year$

Well to Tank (ICE)

$GHG(WTT) = i\sum\{VKTi(km/year) \times FCi\ (l/km) \times (WTT)EFi(kg/l)\}$

$GHG(WTT) = Kg/\ Year$

Grid Emission/ Well to Tank (EV)

$GHG(EV) = i\sum(VKTi \times ECi \times GridEF)$

For electric vehicles, emissions are estimated based on electricity consumption and the carbon intensity of the electricity grid. This approach avoids the common limitation of considering electric vehicles as zero-emission vehicles solely based on tailpipe emissions and provides a more realistic assessment of their environmental performance.

The adoption of a WTW approach aligns the methodology with international lifecycle assessment practices and enables consistent comparison across different vehicle technologies and fuel pathways.

f. Inclusion of Battery Manufacturing Emissions

As part of the lifecycle assessment of electric vehicles, emissions associated with battery manufacturing are also incorporated into the model. Battery production represents a significant source of embodied carbon and can contribute substantially to the overall lifecycle emissions of electric vehicles.

Including battery manufacturing emissions improves the completeness of the assessment by ensuring that comparisons between conventional and electric vehicle technologies account for both operational and upstream impacts. These emissions are annualised over the vehicle lifetime to estimate their contribution to yearly transport system emissions.

Battery Manufacturing (EV)

GHG (B)= Battery Capacity (kWh) x Battery EF (Kg Co2/ kWh) x(Number of vehicle / Vehicle life)

g. Integration of Air Pollutant Emissions and Co-benefits

In addition to greenhouse gas (GHG) emissions, the M3 model incorporates the estimation of key transport-related air pollutants, including Particulate Matter ($PM_{2.5}$), Particulate Matter (PM_{10}), and Nitrogen Oxides (NO_X). These pollutants have significant implications for urban air quality and public health and are therefore considered alongside climate impacts to provide a more comprehensive assessment of transport interventions.

The model adopts a Well-to-Wheel (WTW) approach for pollutant accounting, encompassing both Well-to-Tank (WTT) emissions associated with fuel production, processing, and distribution, and Tank-to-Wheel (TTW) emissions generated during vehicle operation. For electric mobility scenarios, indirect emissions associated with electricity generation are also incorporated where relevant.

Formula for Pollutant Calculation -

● **PM (Total) =** PM (Exhaust) + PM (Non-Exhaust)
● **PM (Exhaust) =** $i\Sigma$ { $VKT_i \times EF\_exhaust\_PM_i$ (Kg/ Km}

● **PM (Non-Exhaust) =** $i\Sigma$ [$VKT_i \times$ {EF(tyre) + EF (brake) + EF(road)}]
● **NOx (Tailpipe)** = $i\Sigma$ { $VKT_i \times EF(NOx_i)$}

h. Overall Justification

The combined ASIF-GREET bottom-up methodology provides a robust framework for assessing the climate impacts of bus-based public transport interventions. By integrating travel activity, modal shifts, vehicle efficiency, fuel characteristics, upstream energy emissions, and future transport scenarios, the methodology captures both existing and potential GHG mitigation benefits at the city level.

The framework supports the assessment of both operational cities and future intervention cities, making it suitable for evaluating current impacts and forecasting future emission-reduction potential. Furthermore, the methodology aligns with internationally recognised transport emission accounting practices while remaining adaptable to local data availability and city-specific conditions.

i. Inferences -

The literature review established four principal findings-

- Bottom-up framework, ASIF framework (Activity X structure X Intensity X Fuel), constitutes the methodological consensus for bus-based urban transport GHG accounting,
- Modal shift from 2W to the public bus transit system represents the highest-yield near-term mitigation lever in Indian cities, with a net emission saving of approximately 0.036 kgCO_2 per passenger-kilometre. In the BBUS project, specifically for the Group B cities, in the scenario development, the modal shift from the 2W will be the highest preference.
- With GHG reduction, there is a need for co benefit analysis in the reduction of NOx, PM 2.5 and PM 10. This is substantial and should be independently quantified and reported alongside carbon metrics.
- Life cycle accounting (Well to Tank +Tank to Wheel) rather than tailpipe-only measurement is required for credible UNFCC reporting, and the BBUS project should report both estimates as a matter of standard practice.
- This calculation will run for one, on an average load factor basis, and a peak hour load factor basis, to understand the increase in GHG emissions and pollutants at different times of the day, and the same type of calculation has been done to understand the seasonal demand fluctuation analysis

4.3 Calculation for Group A Cities

For Group A cities, where bus services are already operational, the methodology is used to estimate the current GHG benefits of public transport. The assessment quantifies the emissions associated with existing travel patterns and compares them with a counterfactual scenario representing travel behaviour in the absence of bus services.

Through passenger surveys and travel behaviour analysis from previous CMP trends (travel behaviour before starting the PM e-Bus Scheme), the model identifies the transport modes displaced by bus usage and estimates the resulting emission reductions. This allows the study to quantify the actual GHG mitigation benefits currently being delivered by bus services.

The GHG reduction by Public transport can be calculated as -

GHG (without Bus)- GHG (Existing) = GHG (Reduced)

Same calculation for PM 2.5, PM 10 and NOx

4.4 Calculation for Group B Cities

For Group B cities, where bus services are currently absent or planned for future implementation, the methodology supports ex-ante assessment of mitigation potential through scenario analysis.

Future emissions are estimated under different development pathways, including these three scenarios-

Scenario	2W Shift	IPT Shift	4W shift	Coverage	Type of procurement
Low Ambition	5%	5%	nil	30%	100% ICE fleet
Moderate ambition	15%	15%	5%	60%	50%ICE+50% EV
High Ambition	25%	20%	15%	80%	100% EV fleet

The difference between the BAU and intervention scenarios represents the potential GHG mitigation that could be achieved through the introduction and expansion of bus services. This enables cities to understand the long-term climate benefits of public transport investments and supports evidence-based planning and decision-making.

Appendix

Appendix A.

A.1 Potential outcomes framework

Let $D_i \in \{0, 1\}$ denote treatment status (1 if household i is within the 500-metre walkshed of a planned/operational bus route, 0 otherwise), and let $t \in \{0, 1\}$ denote survey wave (0 = pre, 1 = post). Define the potential outcome $Y_{it}(d)$ as the outcome that would be observed for household i in wave t under treatment status d. The observed outcome is:

$$Y_{it} = D_i \cdot Y_{it}(1) + (1 - D_i) \cdot Y_{it}(0)$$

The estimand of interest is the Average Treatment Effect on the Treated in the post period:

$$ATT = E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$$

The fundamental problem is that $Y_{i1}(0)$ for treated households is counterfactual. This we can't observe and that's where the DiD mechanism enters.

A.2 Parallel trends and identification

The parallel trends assumption is:

$$E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]$$

That is, in the absence of treatment, the expected change in outcomes for the treated group would have equaled the expected change for the control group. Under parallel trends:

$$ATT = E[Y_{i1} - Y_{i0} \mid D_i = 1] - E[Y_{i1} - Y_{i0} \mid D_i = 0]$$

which is the DiD estimand. Sample analogues yield the standard 2×2 DiD estimator:

$$\widehat{ATT} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre}) = (B - A) - (D - C)$$

where bars denote sample means in the treatment (T) and control (C) groups.

A.3 Heterogeneous treatment effects: triple interaction

The triple-interaction coefficient is β_3.Under conditional parallel trends, β_3 identifies the differential ATT along the VI dimension — the change in the treatment effect for a one-unit increase in vulnerability. The marginal treatment effect at a given vulnerability level v is:

$$\partial ATT/\partial(post \times Bus\_access) \mid VI = v = \beta_2 + \beta_3 \cdot v$$

Appendix B. MECC: Pigouvian Foundation and Derivation

B.1 Private vs social cost on a congested road

Following Walters (1961), let V denote the flow of vehicles on a road segment and C its capacity. The average travel time per vehicle, $t(V)$, is a rising function of flow once a critical threshold is reached. A standard functional form is the Bureau of Public Roads (BPR) function:

$$t(V) = t_0 \cdot [1 + \alpha \cdot (V/C)^\beta]$$

where t_0 is the free-flow travel time, and α, β are empirical parameters ($\alpha \approx 0.15$, $\beta \approx 4$ in the BPR specification). The total travel time on the segment is $V \cdot t(V)$. The marginal private cost faced by a new entrant is $t(V)$ — the additional driver pays for their own travel time but ignores the delay they impose on every other driver. The marginal social cost is:

$$MSC(V) = d/dV\,[V \cdot t(V)] = t(V) + V \cdot t'(V)$$

The wedge $V \cdot t'(V)$ is the marginal external congestion cost — the time imposed on all other users by one additional vehicle. Monetised at the value of time, this is the MECC per vehicle-kilometre.

B.2 From vehicle-km to PCE-km

On a mixed-fleet road network, different vehicle types impose different congestion footprints. The IRC:106 Passenger Car Equivalent (PCE) factor converts a vehicle-kilometre of any vehicle type into car-equivalent units. The relevant network impact of diverting a counterfactual private-vehicle trip is therefore the change in PCE-kilometres, not vehicle-kilometres.

Let m index counterfactual modes (car, two-wheeler, auto, etc.), with d_m the diversion share to public transport, R total bus ridership, occ_m the average occupancy of mode m, L_m the average trip length, and PCE_m the IRC:106 PCE factor. Then:

$$\text{Net PCE-km} = \Sigma_m\,[(R \cdot d_m / occ_m) \cdot L_m \cdot PCE_m] - \text{Bus PCE-km}$$

where the subtraction accounts for the PCE footprint of the bus service itself.

Appendix C. Agglomeration Economies: Production-Function Micro-Foundation

C.1 Production function with effective density

Following Graham (2007) and Venables (2007), assume firms in ward i produce with a Cobb-Douglas production function:

$$y_i = A_i \cdot K_i^{\alpha} \cdot L_i^{(1-\alpha)}$$

where productivity A_i depends on effective economic density U_i:

$$A_i = A_0 \cdot U_i^{\rho}$$

Effective density is the transport-weighted accessibility of employment from i:

$$U_i = \Sigma_j\, E_j \cdot f(d_{ij})$$

with E_j employment in ward j, d_{ij} the generalised travel cost from j to i, and f(·) a distance-decay function. Common specifications are negative-exponential $f(d) = \exp(-\alpha \cdot d)$ and power $f(d) = d^{(-\alpha)}$. The parameter ρ is the agglomeration elasticity — the percentage change in productivity associated with a one-percent change in effective density.

Taking logs:

$$\log y_i = \log A_0 + \rho \cdot \log U_i + \alpha \cdot \log K_i + (1 - \alpha) \cdot \log L_i$$

ρ can in principle be recovered by regressing log productivity on log effective density, controlling for factor inputs. In practice, this is the equation that the international agglomeration literature has estimated and re-estimated, and the identification issues that emerge are central to interpretation.

C.2 Generalized travel cost and agglomeration elasticity

$$d_{ij} = VOT \times (IVT_{ij} + \lambda_w \cdot wait_{ij} + \lambda_a \cdot access_{ij} + \lambda_t \cdot transfers_{ij}) + fare_{ij}$$

The agglomeration elasticity ρ is the single most important parameter in the framework. International estimates from observational wage and productivity regressions range from approximately 0.02–0.05 in advanced-economy settings (Combes, Duranton, and Gobillon 2008; Combes, Duranton, Gobillon, and Roux 2010), with higher and less precisely-bounded values reported for urbanised developing economies (Chauvin et al. 2017). A straightforward import of a UK or French ρ for an Indian application is methodologically uncomfortable — labour-market institutions differ, and the upper bound for India is poorly constrained — but the alternative of estimating ρ from primary data is not practicable within the M1 scope.

We therefore impute ρ from the existing literature, using the value most defensibly transferable to the Indian setting and reporting the WEB estimate as a range rather than a point. The central value used is $\rho = 0.04$, the midpoint of the Combes–Duranton–Gobillon range and broadly consistent with Graham's (2007) UK estimates; sensitivity bounds at $\rho = 0.02$ and $\rho = 0.06$ are reported alongside, capturing both the conservative end of the advanced-economy range and the lower end of the developing-economy values in Chauvin et al. (2017).

Appendix D: Detailed variables list for socio economic benefits and wider economic benefits

Socio-demographic

#	Variable	Unit	source	Role
1	Gender	Categorical (Male/Female/Other)	primary survey	Stratification and control; gender-disaggregated outcomes.
2	Age (years)	Continuous	primary survey	Identifies elderly (60+) for VI; control in all regressions.
3	Household size	Count	primary survey	Control; affects per-capita expenditure and mobility demand.

#	Variable	Unit	source	Role
4	Number of children (<18) in household	Count	primary survey	Educational access sub-sample.
5	Number of elderly (60+) in household	Count	primary survey	Healthcare access sub-sample.
6	Disability status	Binary	primary survey	VI component; key VRU indicator.
7	Household monthly income bracket	Ordinal	primary survey	VI component (normalised, reversed); income decile disaggregation.
8	Household vehicle ownership	Binary	primary survey	VI component.

Transport expenditure

#	Variable	Unit	source	Role
9	Primary transport mode	Categorical	primary survey	Classifies bus users vs non-users for descriptive analysis.
10	Monthly transport expenditure	₹ per month (continuous)	primary survey	Primary dependent variable.
11	Counterfactual expenditure without bus	₹ per month (stated preference)	primary survey	Counterfactual; enables direct computation of bus-attributable savings.
12	Direction of expenditure change (12 months)	Ordinal (categories like increased or decreased)	primary survey	Descriptive trend; cross-validates Group B pre/post.
13	Primary reason for expenditure change	Categorical	primary survey	Attribution check — bus adoption vs fuel price vs distance change.

Employment status

#	Variable	Unit	source	Role
14	Employment status (8 categories)	Categorical; binary recode for regression	primary survey	Primary dependent variable.
15	Commute mode to workplace	Categorical	primary survey	Cross-validates bus usage for employment trips.
16	Distance from home to workplace	Ordinal(distance bands) /continuos(Kms)	primary survey	Captures mobility-enabled labour market access; control.

Women's trip frequency

#	Variable	Unit	source	Role
17	Daily trips outside home (weekday)	Count (integer)	primary survey	Primary dependent variable.
18	Trip purposes (multi-select)	Categorical	primary survey	Disaggregates trip frequency by purpose.
19	Primary transport mode for trips outside home	Categorical	primary survey	Identifies bus-dependent women.
20	Counterfactual: trips made without bus	Count (stated preference)	primary survey	Direct mobility counterfactual.

Educational access

#	Variable	Unit	source	Role
21	Number of children/youth in education in HH	Count	primary survey	Determines sub-sample size.
22	Primary commute mode for children to school/college	Categorical	primary survey	Identifies bus-dependent students.
23	School/college days missed due to transport (3 months)	Count	primary survey	Key descriptive outcome.
24	Counterfactual: would children attend the	Ordinal	primary survey	Captures access to preferred vs nearest institution.

#	Variable	Unit	source	Role
21	Number of children/youth in education in HH	Count	primary survey	Determines sub-sample size.
	same institution without bus?			
25	Change in attendance since bus deployment	Ordinal	primary survey	Self-reported pre/post; Group B.

Healthcare access

#	Variable	Unit	source	Role
26	Number of healthcare visits in last 3 months (HH)	Count	primary survey	Frequency indicator; descriptive outcome.
27	Mode used for healthcare visits	Categorical	primary survey	Identifies bus-dependent healthcare trips.
28	Delayed or skipped healthcare visit due to transport	Binary	primary survey	Unmet healthcare need attributable to transport difficulty.
29	Change in healthcare access since bus deployment	Ordinal	primary survey	Self-reported pre/post; Group B; retrospective Group A.

Spatial / GIS variables

#	Variable	Source	Role
30	Bus access dummy [BusAccess_j]	GIS — MoHUA/STU shapefiles, OSM, GTFS	PRIMARY independent variable. Binary: 1 if a household within 500m walkshed; 0 outside.
31	Walking distance to nearest bus stop (m)	GIS (continuous)	Continuous version for robustness at 400m/800m.
32	Zone classification (treatment/control)	GIS-defined buffer + matched zone assignment	Group B DiD design variable.
33	Population density (ward/zone)	Census 2011 / 2024 ward-level data	Zone-level control; matching variable for Group B.

#	Variable	Source	Role
34	Existing road quality index (ward)	MoRTH / state PWD records	Zone-level control; baseline-infrastructure comparability for matched DiD.

agglomeration framework

Variable	Stage	Unit	source
Ward boundaries	1	—	Municipal corporations / Census 2013
Destination employment E_j	2	Count (jobs)	Economic Census 2013
Distance decay α	2	—	From literature; sensitivity reported
Generalised cost matrix d_ij	2, 3	₹ per trip	Primary survey
Bus deployment scenario	3	Counts of buses, frequency, routes	PM e-Bus Sewa DPR; STU operation plans
Agglomeration elasticity ρ	4	—	imputed from literature
Sectoral GVA-per-worker	4	₹ per worker per year	ASI (MoSPI), NAS, PLFS, NSSO
Ward-level GVA	4	₹ per ward per year	Computed using sectoral employment shares from EC 2013

Appendix E: Detailed variables list for GHG calculation and Emission factors for pollutants

1. **Activity Variables**

S No	Variables		Unit	Type of variable	Data Sources	Importance of data collection
1	Vehicle Kilometres Travelled (VKT)	Primary activity variable	Km/ year	Scope 1	VAHAN	Will be calculated

		Total Veh Reg x survival rate or utilization rate x Annual mileage				
2	Vehicle Occupancy / Load factor	Convert VKT → PKT	pax/ vehicle		ITF India LCA tool defaults	Moderate
3	Vehicle Utilisation Rate		%		assumption/ Research journal	Nil
4	Passenger Kilometre Travelled (PKT)	VKT X Load factor	Pax-km / year		-	Will be calculated
6	Number of vehicles by category	Fleet scale	Count		VAHAN	Nil
7	Average trip length	Validation	Km		Recent CMP/ User survey	High

2. Structure Variables

S No	Variables	Unit	Type of variable	Data Sources	Importance of Primary Survey
1	Modal Share	%	Scope 1	Recent CMP/ OD survey	High
	Vehicle category distribution	%		VAHAN	Nil

2	Fuel mix (Petrol/ diesel/CNG/electric)	%		VAHAN	
3	Technology mix (ICE/HEV/PHEV/BEV/FCEV)	%		VAHAN	
4	Fleet age distribution	%		Vehicle reg (VAHAN) X Survival curve	
5	Public vs Private transport	%		(2w+4w)- Private	Will be calculated

3. **Intensity Variables**

S No	Variables	Unit	Type of variables	Data Sources	Importance of Primary Survey
1	Fuel Consumption (ICE PT/ IPT)	km/l	Scope 1	ARAI	Nil
2	Energy consumption (EV-PT/IPT)	kWh/km		ARAI, Bureau of Energy Efficiency	

Fuel Economy tables for different vehicle types

Vehicle Type	Fuel	Fuel Economy
2W Motorcycle	Petrol	45 km/L

2W Scooter	Petrol	40 km/L
3W Auto	Petrol	20 km/L
3W Auto	Diesel	20 km/L
3W Auto	CNG	25 km/kg
Car	Petrol	16 km/L
Car	Diesel	20 km/L
Car	CNG	24 km/kg
MUV	Diesel	14 km/L
LCV	Diesel	10 km/L
LCV	CNG	8 km/kg
City Bus	Diesel	3 km/L
City Bus	CNG	2.8 km/kg

4. **Fuel factors Variables - TTW (Kg/l) or (Kg/ Km)**

Petrol Carbon factor	2.31 kg CO_2/liter
Diesel Carbon factor	2.68 kg CO_2/liter
CNG Carbon factor	2.75 kg CO_2/kg

WTT & Grid & Battery

ICE	**Petrol Carbon factor**	0.60–0.70 kg/L
	Diesel Carbon factor	0.80 kg/L
	CNG Carbon factor	0.34 kg/kg
EV	**Grid Emission Factor**	0.727 kg/kWh

	Battery production emissions	Kg Co2/kWh (Battery)

5. **PM 10 Emission factor (ARAI, 2025) (g/km)**

Vehicle Type	**BS-II**	**BS-III**	**BS-IV**	**BS-VI**
2W-Motorcycle (Petrol)	0.0075	0.00585	0.00497	0.00367
2W-Scooter (Petrol)	0.0105	0.00975	0.00829	0.00367
3W-CNG	0.177	0.1182	0.03188	0.0125
3W-Diesel	0.65145	0.21045	0.05419	0.02125
3W-LPG	0.177	0.1182	0.03188	0.0125
Car-CNG	0.003	0.00105	0.00098	0.00065
Car-Diesel	0.09	0.04035	0.0231	0.00383
Car-Petrol	0.003	0.00146	0.00098	0.00038
HCV-Bus CNG		0.3672	0.06885	0.02295
HCV-Bus Diesel		0.723	0.0459	0.02295
LCV-CNG			0.0045	0.00383
LCV-Diesel	0.44675	0.181	0.0858	0.00383
MUV-CNG			0.00225	0.00383
MUV-Diesel		0.08655	0.07395	0.00383

6. **PM 2.5 Emission factor (ARAI, 2025) (g/km)**

Vehicle Type	**BS-II**	**BS-III**	**BS-IV**	**BS-VI**
2W-Motorcycle (Petrol)	0.00675	0.00527	0.00448	0.0033

2W-Scooter (Petrol)	0.00945	0.00878	0.00746	0.0033
3W-CNG	0.1593	0.10638	0.02869	0.01125
3W-Diesel	0.58631	0.18941	0.04877	0.01913
3W-LPG	0.1593	0.10638	0.02869	0.01125
Car-CNG	0.0027	0.00095	0.00088	0.00059
Car-Diesel	0.081	0.03632	0.02079	0.00344
Car-Petrol	0.0027	0.00132	0.00088	0.00034
HCV-Bus CNG		0.33048	0.06197	0.02066
HCV-Bus Diesel		0.6507	0.04131	0.02066
LCV-CNG			0.00405	0.00344
LCV-Diesel	0.40208	0.1629	0.07722	0.00344
MUV-CNG			0.00203	0.00344
MUV-Diesel		0.0779	0.06656	0.00344

7. **NOx Emission factor (ARAI, 2025) (g/km)**

Vehicle Type	**BS-II**	**BS-III**	**BS-IV**	**BS-VI**
2W-Motorcycle (Petrol)	0.65825	0.63288	0.53794	0.12033
2W-Scooter (Petrol)	0.5375	0.58	0.493	0.12033
3W-CNG	0.475	2.44825	1.1475	0.15802
3W-Diesel	2.19175	1.7925	0.595	0.19448
3W-LPG	5.15	2.44825	1.1475	0.15802
Car-CNG	2.325	1.96575	0.085	0.085

Car-Diesel	1.225	1.05038	0.54612	0.09724
Car-Petrol	0.225	0.15	0.13375	0.14315
HCV-Bus CNG		10.9395	7.65765	1.00643
HCV-Bus Diesel		12.92875	7.65765	1.00643
LCV-CNG			0.2125	0.15938
LCV-Diesel	4.76542	4.23083	2.4025	0.22313
MUV-CNG			0.407	0.1275
MUV-Diesel		1.6	1.19675	0.17